
The Warehouse Run

The Multi-Robot Amazon Warehouse Exercise,
Recast as an Epistemic Domain and Executed on ePlanSys

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Epistemic Robotics

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The RoboticsAcademy multi-robot Amazon warehouse exercise gives a fleet a floor of racks, a pallet and a dock, and asks for a task planner that assigns the jobs. It never asks how a robot comes to *know* where the pallet is. Here that is the whole problem. The floor is that exercise's own, AWS RoboMaker's small warehouse at its own coordinates, and the plan is not a sequence but a policy, whose branch is decided by what a laser returns.

1 What the exercise leaves out

The exercise is a good one and it is not about knowledge. Two robots are given a warehouse, a pallet and a dock; the student writes something that decides who fetches what, and the robots drive to coordinates they were handed. The uncertainty a real fleet works under has been removed before the exercise begins: somebody already knows where the pallet is, and the planner is that somebody.

Restore the uncertainty and the problem changes shape rather than size. Suppose the pallet is in one of two aisles and no robot knows which. Then there is no sequence of actions that delivers it. Every sequence commits, in advance, to an aisle, and half of them are wrong. What the mission needs is a policy: drive to a place where the question can be answered, answer it, and continue accordingly. Producing that requires a planner whose states are not situations but *sets* of situations an agent cannot tell apart, which is what epistemic planning is for.

Two further things fall out of restoring it, and both are visible in the run.

A precondition about the robot rather than about the warehouse. A robot may lift the pallet only where it *knows* the pallet is. Drop that conjunct and one recovers exactly the classical domain: a robot that happens to be standing in the right aisle lifts a pallet it has no reason to believe is there, and a plan that never looks at anything is valid.

A map is what has been measured. The robot starts at shipping with an empty occupancy grid. The service lane it has to reach is not blocked, it is *unmeasured*, and a route planner that treats unmeasured floor as free plans through walls it has not met. This is the same distinction one level down, and it is answered by the same repository's μ -calculus route planner rather than by the epistemic one.

This report describes the domain **robot-warehouse**, its solution for one, two and three agents, and its execution in Gazebo on ePlanSys with SLAM and Nav2 underneath.

2 The floor

The grid is not a drawing that resembles the exercise's warehouse. It is rasterised from the collision meshes Gazebo itself loads for that world (the building shell, six rack rows, the tall shelf on the west wall and the clutter between them), sliced at the heights a ground robot can strike. Fourteen metres by twenty-one at five centimetres a cell: 281×421 .

The two maps AWS ships beside that world were not used. They are SLAM captures: sparse, noisy, and missing most of the racks, so planning on them would make every result an artefact of whoever drove the robot the day they were recorded.

One detail of that floor decides the shape of the whole domain. The rack rows stop 0.15m short of the east wall. To a planner that treats a robot as a point, that gap is a corridor running the length of the rack block and joining every aisle to every other; routes down it are routes no TurtleBot could drive, since the robot is 0.21m across. Growing the obstacles by the robot's 0.105m radius closes it and leaves 0.7m of each aisle. So in the zone graph an aisle is a *pocket with one mouth*, and there is no edge from one bay to the next. That is why a robot that looks into bay2 and finds it empty has to come back out to the lane, and why the plan pays for that.

3 The domain

Six zones: **dock_south** (shipping) and **dock_north** (receiving) at the two ends of the west **corridor**, the service **lane** along the rack fronts, and **bay2** and **bay3** opening off it. Four propositions and three rigid facts suffice:

(at-ag ?i ?z)	where each robot is
(pallet-at ?z)	which zone the pallet is in
(carrying ?i)	whether ?i has it
(delivered)	whether it has reached a dock
(adjacent ?z1 ?z2), (bay ?z), (dock ?z)	the floor plan, which does not change

Driving, lifting and unloading are *public ontic* actions. A warehouse fleet tracks where its own members are, and the exercise's robots publish odometry to each other, so a move is common knowledge. Keeping movement public is what keeps the pallet the only thing in doubt.

Looking is *semi-private sensing*, with one event per answer:

```
(:event e-inspect-found
:parameters (?i - agent ?z - zone)
:precondition
  (and (at-ag ?i ?z) (bay ?z) (pallet-at ?z) (<Kw. ?i> (pallet-at ?z))))

(:event e-inspect-empty
:parameters (?i - agent ?z - zone)
:precondition
  (and (at-ag ?i ?z) (bay ?z) (not (pallet-at ?z)) (<Kw. ?i> (pallet-at ?z))))

(:action inspect
:parameters (?i - agent ?z - zone)
:action-type (semi-private-sensing
  (e-inspect-found ?i ?z)
  (e-inspect-empty ?i ?z))
:observability-conditions (:and (?i Fully) (default Partially)))
```

Semi-private says that the robot which looks learns the answer, while anyone else present sees it look without learning what it saw. With one robot the distinction is invisible; modelling it as public would nevertheless be a lie, and one that stops being harmless the moment a second robot is added. The `<Kw. ?i>` conjunct in both preconditions says the question must be open: an inspection that settles nothing is not an action, it is a no-op with a cost.

The two events are contradictory, and neither is known to apply when the plan is built. That is exactly what makes the solution branch.

3.1 The precondition the domain exists for

```
(:event e-pickup
:parameters (?i - agent ?z - zone)
:precondition
  (and (at-ag ?i ?z) (bay ?z) (pallet-at ?z) ([?i] (pallet-at ?z))
    (not (carrying ?i))))
:effects (:and (carrying ?i) (not (pallet-at ?z))))
```

(`pallet-at ?z`) is a claim about the warehouse; (`[?i] (pallet-at ?z)`) is a claim about the robot. Both are required. The first makes lifting possible, the second makes it *permitted*, and it is the second that the whole policy is built to establish.

3.2 The instance

The ROS demonstration runs `instances/problem_1.epddl`: one robot, starting at shipping. What the initial state does not say is which bay the pallet is in. It says only that it is in exactly one of them, and says so as common knowledge, so the model designates two worlds rather than one:

```
([C. All] (and (or (pallet-at bay2) (pallet-at bay3))
  (not (and (pallet-at bay2) (pallet-at bay3)))
  ...))
(:forall (?i - agent) ([C. All] (<Kw. ?i> (pallet-at bay2))))
(:forall (?i - agent) ([C. All] (<Kw. ?i> (pallet-at bay3))))

(:goal (and (delivered) ([Kw. r1] (pallet-at bay2))))
```

The goal asks for two things of different kinds at once. (*delivered*) is ontic and could in principle be had by luck. Kw_{r_1} *pallet-at bay₂* cannot: no amount of moving the pallet satisfies a conjunct under Kw , and only coming to know does. That is what forces the plan to sense.

4 Grounding and search

`plank` parses and grounds the instance; ALETHEIA, running as ePlanSys’s plan-solver plugin, searches it.

Atoms	62	(13 of them rigid facts)
Ground actions		66
Initial worlds		2, both designated
Goal modal depth		1
Search depth		11
Nodes expanded		1 008
Time to solve	0.01 s standalone; 1.4 s through the ePlanSys client	

The result is a policy of 17 nodes with a single branch point:

```
go_r1_dock_south_lane [public-ontic]
go_r1_lane_bay2 [public-ontic]
inspect_r1_bay2 [semi-private-sensing]
+-- e-inspect-empty (not pallet-at_bay2)
    go_r1_bay2_lane, go_r1_lane_bay3, pickup_r1_bay3,
    go_r1_bay3_lane, go_r1_lane_dock_south,
    go_r1_dock_south_corridor, go_r1_corridor_dock_north,
    unload_r1_dock_north => goal reached
+-- e-inspect-found (pallet-at_bay2)
    pickup_r1_bay2, go_r1_bay2_lane, go_r1_lane_dock_south,
    go_r1_dock_south_corridor, go_r1_corridor_dock_north,
    unload_r1_dock_north => goal reached
```

Two leaves. A plan with more than one leaf is not a sequence, and which branch runs is decided by what the robot finds rather than by the planner. Note also what the empty branch costs: three extra moves, because there is no edge from *bay2* to *bay3*. The building charges for a wrong guess.

The product update along that policy is reproduced independently of the planner, from the model `plank` exported and the events it ground, and checked at every node: each action the planner used must apply, and the goal must hold at every leaf. The trace and the check are printed by `scenarios/warehouse/tools/show_plan.py`, which exits non-zero if the two disagree.

5 Where the observation comes from

Nothing in the ROS package decides what was seen. `look_into` faces the bay, holds the robot there, and ends when the *model* reports Kw_{r_1} *pallet-at bay₂*. What puts that knowledge in the model is `plansys2_epistemic_perception`, classifying the two bays over the occupancy grid `slam_toolbox` has built from the robot’s own laser, and reporting to the epistemic state which of the two events fired.

If the action node decided it, the demonstration would be a puppet show.

One parameter earns its place here. A bay is visible from the service lane thirty to sixty seconds before the robot reaches it, and the epistemic state will not accept an observation from a robot it does not yet believe is in the bay. Perception therefore keeps re-offering the reading until the executor has applied the drive: `applicability_retries`, set to 240. The default of 40 grids is a few seconds, which fits a building of small rooms and not a warehouse aisle.

6 Crossing a floor that is half unmeasured

`goto_zone` asks the μ -calculus planner for a route and follows it. It does not plan one. That division is the reason for having both planners: ePlanSys decides which zone to be in and what must be known there; the μ -calculus decides how to cross a floor SLAM has only partly filled in, and refuses, while the way there is unobserved, to cross it at all. Early in a run the answer is genuinely *no route*, and the log says so:

```
to lane: at (-3.50, -9.30), asked 'lane', 0 legs; 0 cells of lane measured so far
no route to lane yet; going to look from (-1.96, -9.28)
to lane: at (-1.76, -9.03), asked 'lane', 15 legs, and the zone is not on the map yet
to lane: at (0.36, -8.83), asked 'frontier', 24 legs; 285 cells of lane measured
```

Standing still is not a reply to that, so when the destination comes back unreachable the drive picks a *frontier*: a cell that is known free, has room for the robot and the planner's inflation around it, and has unmeasured floor within sensor range. It routes to that instead, and asks again. Three details are load-bearing, and each of them was, at some point, a robot going nowhere:

- **The target must border the unknown.** Score by closeness to the goal alone and the robot walks to whichever corner of the measured region points at the destination and stops: the map stops growing and the goal stays unreachable.
- **The score rewards travelling.** Subtracting the distance to the frontier over known floor is what gets the robot out of a dead end.
- **A chosen frontier is committed to until it is reached.** A route to a frontier succeeds, which clears the no-route flag, which makes the next cycle ask for the destination, fail, and choose a *different* frontier. Left alone the robot alternates between two of them and travels nowhere.

None of this decides anything epistemic. It is how the robot earns the map that the μ -calculus planner, and then perception, are entitled to reason about.

7 Execution

The world is launched unmodified, a TurtleBot3 burger is spawned at shipping, and the mission runs end to end. Timings are from the run's own log, relative to the first action dispatched:

<i>t</i> (s)	Event	
-2	problem seeded; asking for a plan	
-1	policy: 17 nodes, 1 of them branching	
0	executing <code>goto_zone</code>	shipping → lane → bay 2
79	executing <code>look_into</code>	
79.2	(Kw r1 pallet-at_bay2) holds after 0.4s	the model settles
80	executing <code>pick_up</code>	now permitted
84	executing <code>goto_zone</code>	bay 2 → lane → ... → receiving
212	executing <code>drop_off</code>	
227	mission complete: the policy reached its goal	

The found branch was taken: the pallet was in bay2, so the policy went straight from `look_into` to `pick_up` without the three extra moves the empty branch would have cost.

The recording is 76s at 5×, 3840 × 1040, Gazebo on the left and RViz on the right, captured on a virtual display. The captions are burned in by `ffmpeg` from the run's own log (the actions ePlanSys dispatched and the moment the epistemic state settled), so they cannot drift from what the robot did. The epistemic status in the corner is not narration either: it is a live `EpistemicStateClient::check_formula` query, and it reads UNKNOWN until the robot has looked.

Why the caption is not an RViz marker. An RViz text marker is geometry in the 3-D scene. It scales with the camera, spaces its words oddly, and once a 3840-pixel capture is scaled down for viewing it is a row of thin grey smudges. Several attempts at making it legible that way failed, and they failed for a reason that was not going to be tuned away.

8 What it costs to add a robot

The two-agent instance moves the epistemic conjunct to the robot that never leaves receiving: $delivered \wedge Kw_{r_2} \text{pallet-at bay}_2$. The three-agent instance asks it of r_2 and r_3 , neither of whom goes to look.

Agents	Atoms	Ground actions	Worlds	Depth	Expanded	Generated	Seconds
1	62	66	2	11	1 008	1 012	0.01
2	69	132	2	11	18 254	18 525	0.23
3	76	198	2	12	195 178	197 533	2.90

Nodes expanded grows by roughly an order of magnitude per robot ($\times 18$, then $\times 11$). The *mission* does not get harder: the plan is the same shape and r_1 still does all the walking. The search does. Every agent is another relation over the worlds, and a formula about what somebody knows has to be checked against all of them.

The announcement the planner declines to make. The two-agent policy contains no announcement, although the domain offers `report-pallet-at`. It does not need one: `pickup` is public, so the fleet seeing r_1 lift the pallet out of a bay is already enough for r_2 to know which bay it was. Add a third robot who must also know and the planner does spend an action on it, inserting `report-pallet-at_r1_bay3` on the branch where r_1 found `bay2` empty, because there the knowledge comes from an absence, and watching a lift in `bay3` settles `bay3`, not `bay2`. This is not something the domain was written to demonstrate; it is what the search found.

9 Two rejections

Both are checked by `plank validate`, and both are the expected result.

A sequence, where a policy is needed. Handing the planner the straight-line sequence that assumes the pallet is in `bay2` is rejected: it is unexecutable in the world where the pallet is in `bay3`, which the agent cannot rule out.

Picking up without looking. `pickup_r1_bay2`, after driving to the lane and in to `bay2`, is refused, with `false (pickup_r1_bay2 is not applicable ...)`, because $[r_1] \text{pallet-at bay}_2$ does not hold. Delete that conjunct from `e-pickup` and the same sequence becomes valid, which is exactly the classical domain this one is not.

10 Reproducing

```
# one, two and three agents; the two rejections; the scaling table
bash epddl-workspace/robot-warehouse/validate.sh

# the floor, the six mu-calculus cases, offline and over ROS
colcon build --packages-select epistemic_msgs mu_path_planner warehouse_scenario
bash scenarios/warehouse/run_demo.sh
colcon test --packages-select warehouse_scenario # 12 tests

# the full system in Gazebo
ros2 launch warehouse_demo warehouse_demo_launch.py
```

The domain and its instances are in `epddl-workspace/robot-warehouse`; the floor, its rasteriser and the route-planning cases are in `scenarios/warehouse`; the ROS side, meaning the action nodes, the parameters, the Nav2 configuration and the launch file, is in `ros2_ws/src/warehouse_demo`.